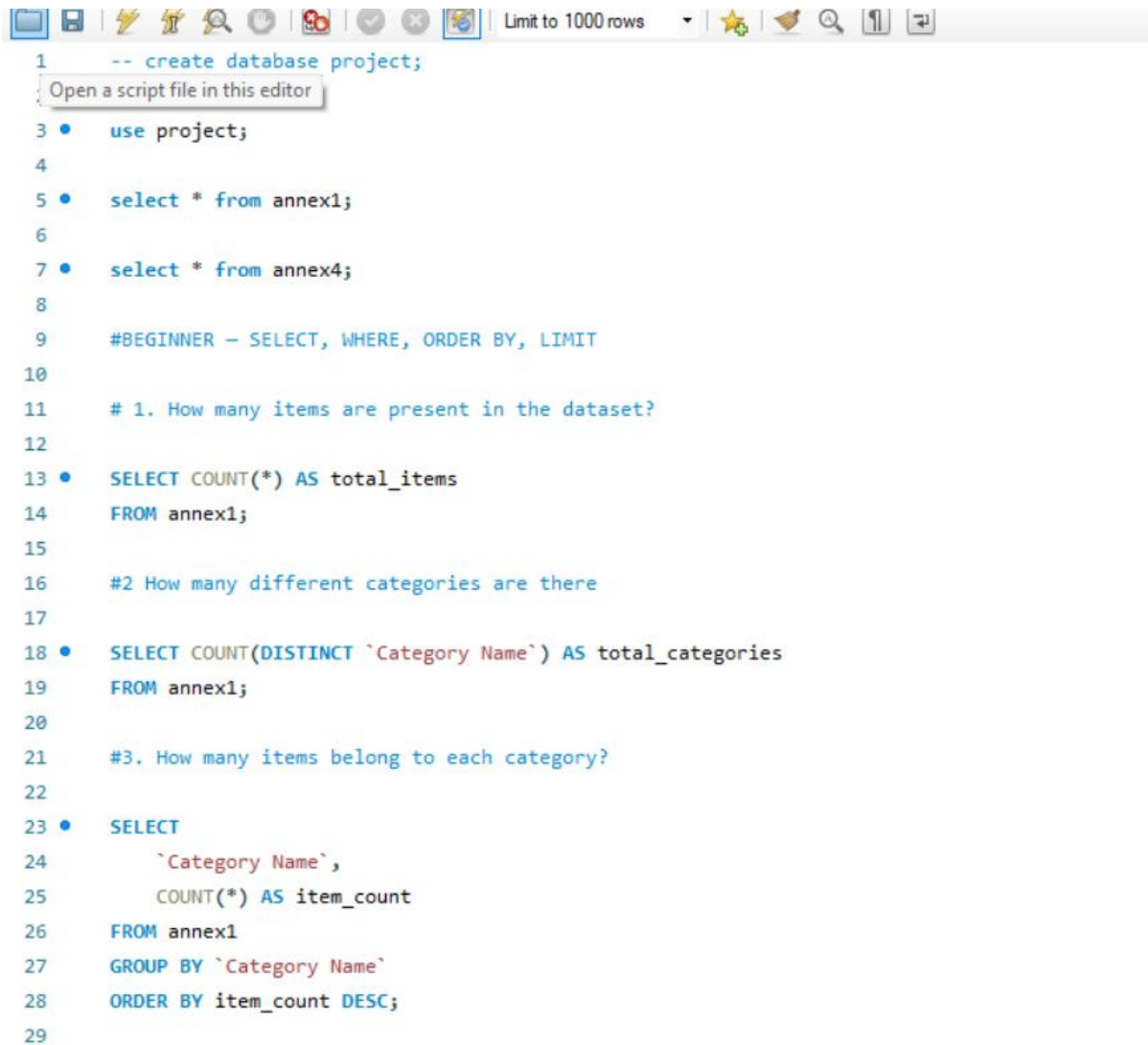




The image shows a screenshot of a SQL editor interface. At the top, there is a toolbar with various icons for file operations, execution, and search. Below the toolbar, a text box prompts the user to "Open a script file in this editor". The main area contains a list of SQL queries, each preceded by a line number and a blue bullet point. The queries are as follows:

```
1  -- create database project;
3  • use project;
4
5  • select * from annex1;
6
7  • select * from annex4;
8
9  #BEGINNER – SELECT, WHERE, ORDER BY, LIMIT
10
11  # 1. How many items are present in the dataset?
12
13  • SELECT COUNT(*) AS total_items
14    FROM annex1;
15
16  #2 How many different categories are there
17
18  • SELECT COUNT(DISTINCT `Category Name`) AS total_categories
19    FROM annex1;
20
21  #3. How many items belong to each category?
22
23  • SELECT
24      `Category Name`,
25      COUNT(*) AS item_count
26    FROM annex1
27   GROUP BY `Category Name`
28   ORDER BY item_count DESC;
29
```

```
31 #4. List all items belonging to the Cabbage category.
```

```
32
```

```
33 • SELECT
```

```
34     `Item Code`,
```

```
35     `Item Name`
```

```
36 FROM annex1
```

```
37 WHERE `Category Name` = 'Cabbage';
```

```
38
```

```
39
```

```
40 #5. Find all items with a loss rate greater than 20%.
```

```
41
```

```
42 • SELECT
```

```
43     `Item Name`,
```

```
44     `Loss Rate (%)`
```

```
45 FROM annex4
```

```
46 WHERE `Loss Rate (%)` > 20
```

```
47 ORDER BY `Loss Rate (%)` DESC;
```

```
48
```

```
49 #6. Find the 10 items with the highest loss rate.
```

```
50
```

```
51 • SELECT
```

```
52     `Item Name`,
```

```
53     `Loss Rate (%)`
```

```
54 FROM annex4
```

```
55 ORDER BY `Loss Rate (%)` DESC
```

```
56 LIMIT 10;
```

```

8 • ALTER TABLE annex4
9   CHANGE `i»çItem Code` `Item Code` BIGINT;
0
1 • DESCRIBE annex4;
2
3 #INTERMEDIATE
4 # 7. Display item name, category and loss rate together.
5
6 • SELECT
7     a.`Item Code`,
8     a.`Item Name`,
9     a.`Category Name`,
0     b.`Loss Rate (%)`
1 FROM annex1 AS a
2 INNER JOIN annex4 AS b
3     ON a.`Item Code` = b.`Item Code`;
4
5 #8. What is the average loss rate of all items?
6
7 • SELECT
8     ROUND(AVG(`Loss Rate (%)`), 2) AS average_loss_rate
9 FROM annex4;
0
1 #9 What is the highest loss rate?
2
3 • SELECT
4     MAX(`Loss Rate (%)`) AS highest_loss_rate
5 FROM annex4;
6

```

```
86
87 #10 Which item has the highest loss rate?
88
89
90 • SELECT
91     `Item Name`,
92     `Loss Rate (%)`
93 FROM annex4
94 ORDER BY `Loss Rate (%)` DESC
95 LIMIT 1;
96
97 #11 What is the lowest loss rate?
98
99 • SELECT
100     MIN(`Loss Rate (%)`) AS lowest_loss_rate
101 FROM annex4;
102
103 #12. Find the average loss rate for each category.
104
105 • SELECT
106     a.`Category Name`,
107     ROUND(AVG(b.`Loss Rate (%)`), 2) AS average_loss_rate
108 FROM annex1 a
109 JOIN annex4 b
110     ON a.`Item Code` = b.`Item Code`
111 GROUP BY a.`Category Name`
112 ORDER BY average_loss_rate DESC;
113
```

```

113
114 #13. Which categories have an average loss rate above 10%?
115
116 • SELECT
117     a.`Category Name`,
118     ROUND(AVG(b.`Loss Rate (%)`), 2) AS average_loss_rate
119 FROM annex1 a
120 JOIN annex4 b
121     ON a.`Item Code` = b.`Item Code`
122 GROUP BY a.`Category Name`
123 HAVING AVG(b.`Loss Rate (%)`) > 10
124 ORDER BY average_loss_rate DESC;
125
126 #14. How many items in each category have a loss rate above 10%?
127
128 • SELECT
129     a.`Category Name`,
130     COUNT(*) AS high_loss_items
131 FROM annex1 a
132 JOIN annex4 b
133     ON a.`Item Code` = b.`Item Code`
134 WHERE b.`Loss Rate (%)` > 10
135 GROUP BY a.`Category Name`
136 ORDER BY high_loss_items DESC;
137
138 #15. Find average, minimum and maximum loss rate for each category.
139
140 • SELECT
141     a.`Category Name`,
142     ROUND(AVG(b.`Loss Rate (%)`), 2) AS average_loss,
143     MIN(b.`Loss Rate (%)`) AS minimum_loss,
144     MAX(b.`Loss Rate (%)`) AS maximum_loss
145 FROM annex1 a
146 JOIN annex4 b
147     ON a.`Item Code` = b.`Item Code`
148 GROUP BY a.`Category Name`
149 ORDER BY average_loss DESC;
150
151 #16. How many items have a loss rate greater than 10%?
152
153 • SELECT
154     COUNT(*) AS items_above_10_percent
155 FROM annex4
156 WHERE `Loss Rate (%)` > 10;
157

```

```

158     #ADVANCED
159     #17. Find items whose loss rate is higher than the overall average.
160
161     • SELECT
162         a.`Item Name`,
163         a.`Category Name`,
164         b.`Loss Rate (%)`
165     FROM annex1 a
166     JOIN annex4 b
167         ON a.`Item Code` = b.`Item Code`
168     WHERE b.`Loss Rate (%)` > (
169         SELECT AVG(`Loss Rate (%)`)
170         FROM annex4
171     )
172     ORDER BY b.`Loss Rate (%)` DESC;
173
73
74     #18. Rank items by loss rate within each category.
75
76     • SELECT
77         a.`Category Name`,
78         a.`Item Name`,
79         b.`Loss Rate (%)`,
80         RANK() OVER (
81             PARTITION BY a.`Category Name`
82             ORDER BY b.`Loss Rate (%)` DESC
83         ) AS loss_rank
84     FROM annex1 a
85     JOIN annex4 b
86         ON a.`Item Code` = b.`Item Code`
87     ORDER BY a.`Category Name`, loss_rank;
88

```

0
9 #19. Find the highest-loss item from each category.
0

```
1 ● WITH ranked_items AS (  
2     SELECT  
3         a.`Category Name`,  
4         a.`Item Name`,  
5         b.`Loss Rate (%)`,  
6         RANK() OVER (  
7             PARTITION BY a.`Category Name`  
8             ORDER BY b.`Loss Rate (%)` DESC  
9         ) AS rnk  
0     FROM annex1 a  
1     JOIN annex4 b  
2         ON a.`Item Code` = b.`Item Code`  
3 )  
4 SELECT  
5     `Category Name`,  
6     `Item Name`,  
7     `Loss Rate (%)`  
8 FROM ranked_items  
9 WHERE rnk = 1;
```

#20 Find the top 3 highest-loss items in every category.

```
WITH ranked_items AS (  
    SELECT  
        a.`Category Name`,  
        a.`Item Name`,  
        b.`Loss Rate (%)`,  
        RANK() OVER (  
            PARTITION BY a.`Category Name`  
            ORDER BY b.`Loss Rate (%)` DESC  
        ) AS rnk  
    FROM annex1 a  
    JOIN annex4 b  
        ON a.`Item Code` = b.`Item Code`  
)  
SELECT  
    `Category Name`,  
    `Item Name`,  
    `Loss Rate (%)`,  
    rnk  
FROM ranked_items  
WHERE rnk <= 3  
ORDER BY `Category Name`, rnk;
```

235 #21. Which categories have a maximum loss rate above 20%?

236

237 • SELECT

238 a.`Category Name`,

239 MAX(b.`Loss Rate (%)`) AS maximum_loss

240 FROM annex1 a

241 JOIN annex4 b

242 ON a.`Item Code` = b.`Item Code`

243 GROUP BY a.`Category Name`

244 HAVING MAX(b.`Loss Rate (%)`) > 20

245 ORDER BY maximum_loss DESC;

246

247 #22. What percentage of items have a loss rate above 10%?

248

249 • SELECT

250 ROUND(

251 SUM(

252 CASE

253 WHEN `Loss Rate (%)` > 10 THEN 1

254 ELSE 0

255 END

256) * 100.0 / COUNT(*),

257 2

258) AS percentage_high_loss

259 FROM annex4;

260

2

1 #23. Categorize items according to their loss rate.

2

3 • SELECT

4 a.`Item Name`,

5 a.`Category Name`,

6 b.`Loss Rate (%)`,

7 CASE

8 WHEN b.`Loss Rate (%)` = 0 THEN 'No Loss'

9 WHEN b.`Loss Rate (%)` <= 5 THEN 'Low Loss'

10 WHEN b.`Loss Rate (%)` <= 10 THEN 'Medium Loss'

11 WHEN b.`Loss Rate (%)` <= 20 THEN 'High Loss'

12 ELSE 'Very High Loss'

13 END AS loss_category

14 FROM annex1 a

15 JOIN annex4 b

16 ON a.`Item Code` = b.`Item Code`

17 ORDER BY b.`Loss Rate (%)` DESC;

18

```
279 #VIEW
280
281 #24. Create a summary view
282
283 • CREATE VIEW item_loss_summary AS
284 SELECT
285     a.`Item Code`,
286     a.`Item Name`,
287     a.`Category Code`,
288     a.`Category Name`,
289     b.`Loss Rate (%)`
290 FROM annex1 a
291 JOIN annex4 b
292     ON a.`Item Code` = b.`Item Code`;
293
294 #25. Create a category-level summary view
295
296 • CREATE VIEW category_loss_summary AS
297 SELECT
298     a.`Category Name`,
299     COUNT(*) AS total_items,
300     ROUND(AVG(b.`Loss Rate (%)`), 2) AS average_loss,
301     MIN(b.`Loss Rate (%)`) AS minimum_loss,
302     MAX(b.`Loss Rate (%)`) AS maximum_loss
303 FROM annex1 a
304 JOIN annex4 b
305     ON a.`Item Code` = b.`Item Code`
306 GROUP BY a.`Category Name`;
```